# Johnny Kress

Email: Johnnykress808@gmail.com | Phone: (808)-729-2366 | Location: Seattle, WA

## EDUCATION

**University Of Washington** **Expected Graduation, August 2027**

*Bachelor's of Science in Statistics*

Cumulative GPA: 3.77

Relevant Coursework: Statistical Methods & Computing, Data Management, Probability 1, Advanced Linear Algebra, Multivariable Calculus, Computer Programming 1 & 2

## PROFESSIONAL EXPERIENCE

**SeattleStatGROWS Intern**

*Fred Hutch Cancer Research Center | Seattle, WA | Jun 2026 – Aug 2026*

- Conducted a mediation analysis on single-nuclei RNA-seq data from 100 post-mortem donors (558,991 microglia across 7 functional states) to test whether brain immune cell composition mediates the effect of APOE genotype on Alzheimer's disease pathology
- Identified a lipid-processing microglia state as the leading candidate mediator (95% CI: −0.03 to +0.38), a result consistent with the known interaction between amyloid beta and lipids
- Collaborated with the Jayadev & Prater labs (UW Neurology) and presented findings to faculty and peers at the SeattleStatGROWS 2026 research symposium

**Licensed Assistant**

*Windermere R.E. | Shoreline, WA | May 2025 – Present*

- Analyze recent sales data, active listings, and market conditions to advise clients on optimal pricing strategies
- Maintain and update client databases of 200+
- Researched and presented comprehensive reports on commercial development trends and future outlook in Shoreline, WA

## Technical Skills/Projects

**Languages: SQL, Java, Python, R/Rstudio, LaTeX**

**Projects:**

**Robust Mediation Simulation Study in R (Fred Hutch):**

- Built a Monte Carlo simulation framework in R to stress-test the ROBMED package across a 17-scenario grid varying sample size, binary vs. continuous predictors, mediator correlation, and outlier contamination
- Showed that bad-leverage contamination collapses confidence-interval coverage from 95% to 6% while making mediator rankings appear more accurate — a failure mode invisible from point estimates alone
- Calibrated a power curve from 140 simulated mediator conditions and used it to forecast, in advance, which effects were detectable in the real 100-donor dataset and which were underpowered

**K-Means Clustering Algorithm Development in R:**

- Built a k-means clustering algorithm in R from scratch, implementing centroid assignment, iterative optimization, and convergence detection without using built-in functions
- Evaluated clustering quality using WSS elbow method and stability analysis across multiple random initializations to ensure robust model performance
- Simulated datasets with varying cluster structures to test algorithm effectiveness under different data conditions and identify best use cases

## Additional Work/Team Experience

**UW Kendama Club Leader**

- Organized club meetings, taught new players, and coordinated a trip to Japan of 11 group members to compete in the Kendama World Championship
- Led a fundraiser raising $2000+ for the Washington Immigrant Solidarity Network